

JPE Replication Report

PUBLISHED
March 17, 2026

1 Paper Identification

ID	20240145
Author	Miller
Title	Who Values Democracy?
Iteration	1

Some useful information up front:

- [JPE Data and Code Availability Policy](#)
- [Step by step guidance from Data Editor for Authors](#)
- [Template README](#)

2 Summary

In assessing compliance with our [Data and Code Availability Policy](#), our reproducibility team has identified the following issues, which we ask you to address:

1. Please provide a step by step section for replicators. Your scons approach is good, but we hit a bug because of a missing file. You can keep the scons pipeline but you must explain better what requirements there are to run this. Ideally, you could provide an outline of how one could run this package manually - I am sorry about this extra layer, but our experience has shown that this is quite fragile. What's missing:
 1. pdflatex dependency
 2. there is a missing file (see below)
 3. Your setup assumes (via `source/lib/JMSLab/builders/executables.yml`) that the stata executable is `StataMP`. This is not a good assumption, as on our system it's called `stata-mp`. You need to alert the user to those issues as they will create difficult to track failures.
2. There are missign tick boxes below, please add to reamde.

3 General

- ☐ Data Availability and Provenance Statements
 - ☐ Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
 - ☐ Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
 - ☐ License for Data (optional, but recommended)
 - ☒ Details on each Data Source
- ☒ Dataset list
- ☒ Computational requirements
 - ☒ Software Requirements
 - ☐ Controlled Randomness (as necessary)
 - ☒ Memory, Runtime, and Storage Requirements
- ☒ Description of programs/code
 - ☒ License for Code (Optional, but recommended)
- ☒ List of tables and programs
- ☐ References (Optional, but recommended)

4 High-level Description of Research Project

This research project...

- ☒ uses observational data.
- ☒ uses *confidential* observational data.
- ☐ uses data generated via experiment or trial by the authors (in field or lab).
 - ☐ For lab experiments: The code to implement the experimental protocol (z-Tree, o-Tree etc) is included in the package.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF document outlining the design of the experiment, including information on the selection and eligibility of participants is included.
 - ☐ A PDF copy of the instructions given to participants in both original language and an English translation is included.
- ☐ uses data generated via survey by the authors (in field or online)
 - ☐ The programs used to administer the survey (e.g. qualtrics survey file) are included.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF of the original questionnaire(s) and information on the selection and eligibility of participants is included.
- ☐ is a simulation study: the data are generated by an algorithm.
- ☐ is a numerical illustration: no data are used but code is used to produce illustrative output (tables, figures)

5 Data description

For any data that cannot be provided as part of the replication package, please provide the following information as part of the README.

{REQUIRED} Please specify how long the data will be preserved in the restricted-access location.

{REQUIRED} Please provide affirmation of support for replication checks. – As per the [JPE policy](#), “In cases where data cannot be published in an openly accessible trusted data repository like the JPE dataverse, authors who have requested an exemption to publish them at the time of first submission must commit to preserving the data and code for a period of no less than five years following the publication of the paper. They should also provide reasonable assistance to requests for clarification and replication.”

{REQUIRED} Please provide a codebook for the data. – The codebook should at a minimum describe the variables obtained from the data source, in a manner that will let others verify that they have obtained a substantially similar dataset if successfully obtaining access to the data.

5.1 Data Sources

Input data sources used to build the analysis data.

Data Source	Location	Dataset	Public	Private	Access Described	DOI / URL
Correlates of War (CoW)	datastore/raw/cow/	militarized_interstate_disputes/MIDB_4.2.csv	Yes	No	Yes	Yes
Correlates of War (CoW)	datastore/raw/cow/	militarized_interstate_disputes/MIDLOC1_2.0.csv	Yes	No	Yes	Yes
Correlates of War (CoW)	datastore/raw/cow/	wars/Inter-StateWarData_v4.0.dta	Yes	No	Yes	Yes
Correlates of War (CoW)	datastore/raw/cow/	wars/Extra-StateWarData_v4.0.dta	Yes	No	Yes	Yes
Correlates of War (CoW)	datastore/raw/cow/	wars/Intra-StateWarData_v4.1.dta	Yes	No	Yes	Yes

Data Source	Location	Dataset	Public	Private	Access Described	DOI / URL
Fraser Institute's Economic Freedom Index (EFI)	datastore/raw/economic_freedom/	efi_data.dta	Yes	No	Yes	Yes
Global Financial Data (GFD/Finaeon)	datastore/raw/gfd/	country_series/	No	Yes	Yes	Yes
Global Financial Data (GFD/Finaeon)	datastore/raw/gfd/	gfd_data_series_list.xlsx	No	Yes	Yes	Yes
Global Financial Data (GFD/Finaeon)	datastore/raw/gfd/	gfd_recessions.csv	No	Yes	Yes	Yes
Global Financial Data (GFD/Finaeon)	datastore/raw/gfd/	govt_rev_gdp.dta	No	Yes	Yes	Yes
Global Financial Data (GFD/Finaeon)	datastore/raw/gfd/	supplements/extraTotalReturnsClean.xlsx	No	Yes	Yes	Yes
Global Financial Data (GFD/Finaeon)	datastore/raw/gfd/	supplements/price_earnings_clean.csv	No	Yes	Yes	Yes
International Crisis Behavior (ICB)	datastore/raw/icb/	icb2v12.csv	Yes	No	Yes	Yes
Jordà- Schularick- Taylor (JST) Macrohistory Database	datastore/raw/jst/	JSTdatasetR5.dta	Yes	No	Yes	Yes
Maddison Historical Statistics	datastore/raw/ggdc/	mpd2020.dta	Yes	No	Yes	Yes
Penn World Table	datastore/raw/pwt/	pwt100.dta	Yes	No	Yes	Yes
Penn World Table	datastore/raw/pwt/	pwt100-capital-detail.dta	Yes	No	Yes	Yes
Penn World Table	datastore/raw/pwt/	pwt100-labor-detail.dta	Yes	No	Yes	Yes
Relative Political Capacity Dataset (RPC)	datastore/raw/arpc/	arpc_2020_comp.dta	Yes	No	Yes	Yes

Data Source	Location	Dataset	Public	Private	Access Described	DOI / URL
Standardized World Income Inequality Database (SWIID)	datastore/raw/swiid/	swiid9_0_summary.csv	Yes	No	Yes	Yes
Varieties of Democracy (V-Dem)	datastore/raw/vdem/	V-Dem-CY+Others-v8.dta	Yes	No	Yes	Yes
Varieties of Democracy (V-Dem)	datastore/raw/vdem/	V-Dem-CY-Full+Others-v10.dta	Yes	No	Yes	Yes
Wharton Research Data Services (WRDS)	datastore/raw/wrds/	factset_annual_fiscal.dta	No	Yes	Yes	Yes
Wharton Research Data Services (WRDS)	datastore/raw/wrds/	ibes_global_actual.dta	No	Yes	Yes	Yes
World Religion Dataset (WRD)	datastore/raw/cow/religion/	WRP_national.csv	Yes	No	Yes	Yes
Data from Published Papers and Other Events	datastore/raw/other_events/	reinhart_rogooff_defaults.dta	Yes	No	Yes	Yes
Data from Published Papers and Other Events	datastore/raw/other_events/	reinhart_rogooff_financial_crisis.csv	Yes	No	Yes	Yes
Data from Published Papers and Other Events	datastore/raw/other_events/	assassinations_data.dta	Yes	No	Yes	Yes
Data from Published Papers and Other Events	datastore/raw/other_events/	HOG_death.csv	Yes	No	Yes	Yes
Data from Published Papers and Other Events	datastore/raw/other_events/	country_names_to_iso3.dta	Yes	No	Yes	Yes
Country Code Crosswalks	datastore/raw/crosswalks/	country_name_to_iso3.csv	Yes	No	Yes	Yes
Country Code Crosswalks	datastore/raw/crosswalks/	cow_country_codes_to_iso3.csv	Yes	No	Yes	Yes

Data Source	Location	Dataset	Public	Private	Access Described	DOI / URL
Country Code Crosswalks	datastore/raw/crosswalks/	ibes_country_to_iso3_codes.csv	Yes	No	Yes	Yes
Country Code Crosswalks	datastore/raw/crosswalks/	icb_country_code_to_iso3.csv	Yes	No	Yes	Yes
Country Code Crosswalks	datastore/raw/crosswalks/	iso2_codes_to_iso3_codes.csv	Yes	No	Yes	Yes
Country Code Crosswalks	datastore/raw/crosswalks/	region_map.dta	Yes	No	Yes	Yes
Country Code Crosswalks	datastore/raw/crosswalks/	valid_iso3_codes.dta	Yes	No	Yes	Yes
Foreign Direct Investment (FDI)	datastore/raw/fdi/	fdi_inflows_raw.csv	Yes	No	Yes	Yes
Foreign Direct Investment (FDI)	datastore/raw/fdi/	fdi_outflows_raw.csv	Yes	No	Yes	Yes
Regime Change Data	datastore/raw/regime_change/	DDCGdata_final.dta	Yes	No	Yes	Yes
Regime Change Data	datastore/raw/regime_change/	ERT.csv	Yes	No	Yes	Yes
Regime Change Data	datastore/raw/regime_change/	historical_events_for_democratization.csv	Yes	No	Yes	Yes
Regime Change Data	datastore/raw/regime_change/	vod_democratizations.csv	Yes	No	Yes	Yes
World Income Inequality Database (WIID)	datastore/raw/wiid/	wiidcountry.dta	Yes	No	Yes	Yes

6 Requirements

- ☒ README is in TXT, MD, or PDF format
- ☒ README is accessible at the root of the package (not inside a folder)
- ☐ Title conforms to guidance (starts with “Data and Code for: Title of Paper” or “Code for: Title of Paper”, is properly capitalized)
- ☐ Authors (with affiliations) are listed in the same order as on the paper
- {REQUIRED} Please review the title of the deposit as per our guidelines.

{REQUIRED} Please review authors and affiliations on the deposit. In general, they are the same, and in the same order, as for the manuscript; however, authors can deviate from that order.

6.1 Stated Requirements

``The code was last run on a 10-core Intel-based laptop with WindowsOS version 11 Enterprise with 50GB of free space. Computation took 21 minutes”

- ✓ Software Requirements specified as follows:
 - Python >=3.11, <3.13
 - Stata (code was last run with version 19)
- ✓ Computational Requirements specified as follows:
 - 2 GB - 25 GB required storage space
- ✓ Time Requirements specified as follows:
 - 10-60 minutes

7 Analyzing Data and Code Files

7.1 Code description

The author implements the replication using a SCons-based reproducible research pipeline, orchestrated through uv, which automatically runs the Python and Stata scripts needed to generate the paper’s tables and results.

There are 36 `.do` scripts which generate the analysis data.

There are 10 `.do` and 2 `.py` scripts which generate the main tables.

There are 7 `.do` and 1 `.py` scripts which generate the main figures.

There are 16 `.do` and 1 `.py` scripts which generate the appendix tables.

There are 11 `.do` scripts which generate the appendix figures.

Summarizing the Data Cleaning Code

source/derived/01_events/

Dataset Produced	Program
	01_create_valid_iso3_codes.do
	02_create_pre1900_dem.do
	03_create_regime_change.do
	04_create_coup_detat.do
	05_create_financial_crises.do
	06_create_sovereign_default.do
	07_create_hog_deaths.do
	08_create_international_crises.do
	09_create_recession.do
	10_create_interstate_wars.do
	11_create_intrastate_wars.do
	12_create_extrastate_wars.do
	13_create_militarized_disputes.do
	14_combine_all_events.do

source/derived/02_macro_political/

Dataset Produced	Program
	01_create_inflation.do
	02_create_pwt_data.do
	03_create_gdp.do
	04_create_vdem_datasets.do
	05_create_portion_catholic.do
	06_create_gini.do
	07_create_tax_rates.do
	08_create_gov_rev.do
	09_create_economic_competition.do
	10_create_fdi.do

source/derived/03_assets/

Dataset Produced	Program
	01_factset_clean.do
	02_ibes_global_clean.do
	03_create_dividend_yields.do
	04_create_equity_returns.do
	05_create_fixed_income.do
	06_create_home_country_bond_rate.do
	07_create_excess_and_abnormal_returns.do
	08_create_cashflow_growth.do
	09_create_other_asset_market_variables.do
	10_create_return_news.do

source/derived/04_merging/

Dataset Produced	Program
	01_create_section_3_and_5_data.do
	02_create_section_4_data.do

Summarizing code producing Tables and Figures:

If a table or figure is indicated as reproduced with no additional comments, this means it was exactly reproduced.

Figure/Table #	Program	Filename	Reproduced?
Main Tables	source/analysis/main/tables	source/tables/raw	
Table 1	01_table_1_change_in_log_dividend_yield.do	table_1_change_in_log_dividend_yields.tex	Yes - one tiny SE mismatch 5.83 vs 5.82

Figure/Table #	Program	Filename	Reproduced?
Table 2	02_table_2_cashflow_growth.do	table_2_cashflow_growth.tex	Yes - one tiny SE mismatch 1.79 vs 1.80
Table 3	03_table_3_democratization_vs_other_political_risk.do	table_3_democratization_vs_other_political_risk.tex	Yes - 1 small p value mismatch 0.68 vs 0.7 and 7 small SE mismatches 6.71 vs 6.73, 8.70 vs 8.71, 9.25 vs 9.28, 10.60 vs 10.61, 10.81 vs 10.83, 7.88 vs 7.89, 5.00 vs 5.01
Table 4	04_table_4_revolution_risk.do	table_4_revolution_risk.tex	Yes - small SE mismatches 10.57 vs 10.58, 8.39 vs 8.40
Table 5	05_table_5_regional_waves_instrument.do	table_5_regional_waves_instrument.tex	No
Table 6	06_table_6_balance_tests.do	table_6_balance_tests.tex	Yes
Table 7	07_table_7_did_results.do	table_7_did_results.tex	Yes - small SE mismatches 2.24 vs 2.25, 2.21 vs 2.34
Table 8	08_table_8_explicit_redistribution.do	table_8_explicit_redistribution.tex	Yes
Table 9	09_table_9_implicit_redistribution.do	table_9_implicit_redistribution.tex	Yes
Table 10	10_table_10_high_vs_low_redistribution_risk.do	table_10_high_vs_low_redistribution_risk.tex	Yes - small SE mismatches 8.89 vs 8.90, 8.80 vs 8.82
Table 11	11_table_11_model_calibration.py	table_11_model_calibration.tex	No
Table 12	12_table_12_model_results.py	table_12_model_results.tex	No
Main Figures	source/analysis/main/figures	source/figures/raw	
Figure 1	01_figure_1_dividend_yield_event_study.do	figure_1_dividend_yield_event_study.pdf	Yes
Figure 2	02_figure_2_physical_human_capital.do	figure_2_physical_human_capital.pdf	Yes
Figure 3	03_figure_3_gdp_consumption_distributions.do	figure_3a_gdp_growth_distribution.pdf, figure_3b_consumption_growth_distribution.pdf	Partly - Figure 3b's legend partly impedes

Main			
Figures	source/analysis/main/figures	source/figures/raw	
			on the histogram.
Figure 4	04_figure_4_regional_waves.do	figure_4_regional_waves.pdf	Yes
Figure 5	05_figure_5_anti_regime_event_study.do	figure_5a_event_study_anti_system_cso.pdf, figure_5b_event_study_democratic_protests.pdf	yes - the last character of the x axis label is cropped out
Figure 6	06_figure_6_returns_event_study.do	figure_6_did_event_study_returns.pdf	Yes
Figure 7	07_figure_7_dividend_yield_coefficients_over_time.do	figure_7_coefficients_over_time.pdf	Yes
Figure 8	08_figure_8_autocratization_figure.py	figure_8_autocratization_model.pdf	No
Appendix			
Tables	source/analysis/appendix/tables	source/tables/raw	
Table A1	01_table_A1_summary_statistics.do	A1_summary_statistics.tex	Yes
Table B2	02_table_B2_adverse_dividend_growth.do	B2_adverse_dividend_growth.tex	Partly, yes - 6 small mismatches in latec output vs. pdf SEs: 20.08 vs. 20.09; 16.16 vs 16.19; 11.30 vs 11.31; 15.14 vs 15.15; 7.58 vs. 7.61; 7.49 vs. 7.51
Table B3	03_table_B3_risk_premium.do	B3_risk_premium.tex	Yes - small SE mismatches 1.78 vs 1.76, 2.41 vs 2.40, 2.02 vs 2.01, 2.04 vs 2.03, 2.94 vs 2.93, 3.35 vs 3.34, 2.84 vs 2.83, -1.50 vs -1.49, 2.44 vs 2.43, 2.14 vs 2.13, 2.09 vs 2.08, 3.15 vs 3.14, 2.14 vs 2.13, 1.79 vs 1.77, 0.85 vs 0.84
Table B4	04_table_B4_macro_political_risk_measures.do	B4_macro_political_risk_measures.tex	Yes
Table B5	05_table_B5_adverse_probability.do	B5_adverse_probability.tex	Yes - note there is another B5 table generated, which does not

Appendix			
Tables	source/analysis/appendix/tables	source/tables/raw	
			correspond, see screenshot
Table B6	06_table_B6_democratize_risk_measures.do	B6_democratize_risk_measures.tex	Yes
Table C7	07_table_C7_probability_democratize_post_vatican_ii.do	C7_vatican_ii_democratize.tex	Yes - small SE mismatches 0.81 vs 0.83, 1.76 vs 1.88
Table C8	08_table_C8_vatican_i.do	C8_did_first_vatican.tex	Yes - small SE mismatches 3.24 vs 3.29, 2.70 vs 2.74
Table C9	09_table_C9_catholic_democracies.do	C9_did_democracies.tex	Yes - one tiny SE mismatch 5.76 vs. 5.75
Table C10	10_table_C10_removing_outliers.do	C10_did_removing_outliers.tex, C10_did_removing_outliers_long.tex	Yes - some small SE mismatches 3.35 vs 3.36, 3.17 vs 3.18, 1.97 vs 2.01, 1.79 vs 1.81, 2.68 vs 2.69, 2.53 vs 2.67
Table C11	11_table_C11_outlier_robust_weights.do	C11_did_robust_weights.tex	Yes
Table C12	12_table_C12_global_capm.do	C12_did_global_capm.tex	Yes - one small SE mismatch 1.91 vs 1.93
Table C13	13_table_C13_no_rolling_beta.do	C13_did_no_rolling_beta.tex	Yes - one tiny SE mismatch 6.17 vs 6.18
Table C14	14_table_C14_home_country_bonds.do	C14_did_results_country_bonds.tex	Yes
Table C15	15_table_C15_capital_gains_control.do	C15_did_capital_gains.tex	Yes - one tiny SE mismatch 3.78 vs 3.77
Table D16	16_table_D16_inequality_price_decline.do	D16_inequality_price_decline.tex	Yes
Table G17	17_table_G17_ert_democracy.py	table_G17_democratizations_and_history.tex	Yes
Appendix			
Figures	source/analysis/appendix/figures	source/figures/raw	
Figure B1	01_figure_B1_democracy_log_price.do	B1_democracy_log_price_dividend.pdf, B1_democracy_log_price_gdppc.pdf, B1_democracy_log_price_price.pdf	Yes
Figure C2	04_figure_C4_cso_activity_vs_mobilizations.do	C2_democracy_activity_mobil.pdf	No
Figure C3	07_figure_C7_country_pair_1946_1976.do	C3_country_pair_falsification_aut.pdf, C3_country_pair_falsification.pdf	No

Appendix
Figures

source/analysis/appendix/figures

source/figures/raw

Figure C4	06_figure_C6_window_end_date.do	C4_window_end_date_all.pdf, C4_window_end_date_aut.pdf	No
Figure C5	08_figure_C8_country_pair_1939_1983.do	C5_country_pair_PE_short_aut.pdf, C5_country_pair_PE_short.pdf, C5_country_pair_T_short_aut.pdf, C5_country_pair_T_short.pdf	No
Figure C6	08_figure_C8_country_pair_1939_1983.do	C6_country_pair_PE_long_aut.pdf, C6_country_pair_PE_long.pdf, C6_country_pair_T_long_aut.pdf, C6_country_pair_T_long.pdf	No
Figure C7	09_figure_C9_dividend_yield_event_study.do	C7_dividend_yield_event_study.pdf	No
Figure D8	10_figure_D10_explicit_redistribute_event_study.do	D8_explicit_redistribute_event_study_govt_rev_gdp.pdf, D8_explicit_redistribute_event_study_swiid.pdf	No
Figure D9	11_figure_D11_democratization_end_price_response.do	D9_democratize_price_response_ld.pdf, D9_democratize_price_response_succ.pdf	No
Figure F11	12_figure_F13_sweden_case_study.do	F11_sweden_dividend_yield.pdf	No
Figure F12	13_figure_F14_france_case_study.do	F12_france_dividend_yield.pdf	No

Summarizing In-text Numbers

In-text numbers	Location in Manuscript	Program	Line Number	Reproduced?
NOTE: In-text numbers that reference numbers in tables do not need to be listed. Only in-text numbers that correspond to no table or figure need to be listed.				

7.2 Computing Environment of the Replicator

- Mac Laptop M2 MacBook Air, MacOS 15.7.4, 16 GB of memory, (M2 Processor, 8 cores)
- Stata/SE 19.5.0
- VSCode 1.109.2 with Py3.13
- R 4.5.2 with RStudio 2026.01 with Quarto and JDK 21

8 Replication

Ran the following commands in bash to start the replication:

```
conda create -n Miller python=3.12
conda activate Miller
python -m pip install uv
python -m uv run scons
```

The below error was returned after the last command:

```
(Miller) nuvolos@nuvolos:/files/JPE-Miller-20240145/replication-package/replication_package$ python -m uv run scons
scons: Reading SConscript files ...
scons: done reading SConscript files.
scons: Building targets ...
build_python(["source/paper/numbers/section_3.tex"], ["source/analysis/numbers/02_section_3_numbers.py",
"empirics_numbers.json"])
scons: *** [source/paper/numbers/section_3.tex] ExecCallError : Python did not run successfully. Please check that the
```

executable, source, and target files are correctly specified. Check source/paper/numbers/sconscript.log and sconstruct.log for errors.

Command tried: "python" source/analysis/numbers/02_section_3_numbers.py > source/paper/numbers/sconscript.log

Traceback (most recent call last):

```
File "/files/JPE-Miller-20240145/replication-
package/replication_package/source/analysis/numbers/02_section_3_numbers.py", line 11, in <module>
    from source.config import ROOT_PATH
File "/files/JPE-Miller-20240145/replication-package/replication_package/source/config.py", line 10, in <module>
    ROOT_PATH = Path(os.environ["ROOT_PATH"])
                    ~~~~~~
File "<frozen os>", line 685, in __getitem__
KeyError: 'ROOT_PATH'
```

Traceback (most recent call last):

```
File "/files/JPE-Miller-20240145/replication-package/replication_package/.venv/lib/python3.12/site-
packages/SCons/Action.py", line 1441, in execute
    result = self.execfunction(target=target, source=rresources, env=env)
              ~~~~~~
File "/files/JPE-Miller-20240145/replication-package/replication_package/source/lib/JMSLab/builders/build_python.py",
line 28, in build_python
    builder.execute_system_call()
File "/files/JPE-Miller-20240145/replication-package/replication_package/source/lib/JMSLab/builders/jmslab_builder.py",
line 120, in execute_system_call
    self.do_call()
File "/files/JPE-Miller-20240145/replication-package/replication_package/source/lib/JMSLab/builders/jmslab_builder.py",
line 159, in do_call
    self.raise_system_call_exception(traceback = traceback)
File "/files/JPE-Miller-20240145/replication-package/replication_package/source/lib/JMSLab/builders/jmslab_builder.py",
line 186, in raise_system_call_exception
    raise ExecCallError(message)
```

source.lib.JMSLab.exception_classes.ExecCallError: Python did not run successfully. Please check that the executable, source, and target files are correctly specified. Check source/paper/numbers/sconscript.log and sconstruct.log for errors.

Command tried: "python" source/analysis/numbers/02_section_3_numbers.py > source/paper/numbers/sconscript.log

Traceback (most recent call last):

```
File "/files/JPE-Miller-20240145/replication-
package/replication_package/source/analysis/numbers/02_section_3_numbers.py", line 11, in <module>
    from source.config import ROOT_PATH
File "/files/JPE-Miller-20240145/replication-package/replication_package/source/config.py", line 10, in <module>
    ROOT_PATH = Path(os.environ["ROOT_PATH"])
                    ~~~~~~
File "<frozen os>", line 685, in __getitem__
KeyError: 'ROOT_PATH'
```

scons: building terminated because of errors.

Which was fixed using:

```
export ROOT_PATH="$PWD"
python -m uv run scons
```

Which then returned the below error:

```
(Miller) nuvolos@nuvolos:/files/JPE-Miller-20240145/replication-package/replication_package$ python -m uv run scons
scons: Reading SConscript files ...
```

```
scons: *** missing SConscript file 'source/analysis/paper/SConscript'
File "/files/JPE-Miller-20240145/replication-package/replication_package/SConstruct", line 34, in <module>
```

`paper/` is stored in `source/`, not under `source/analysis/`.

The replicator copies the the file and proceeds:

```
mkdir -p source/analysis/paper
cp -r source/paper/* source/analysis/paper/
```

The below error is returned. The build calls `pdfplatex` but this is not listed in the computational requirements, nor included in the dependencies downloaded by the package. Replicator installs the package with:

```
conda install -c conda-forge texlive-core
```

and proceeds:

```
(Miller) nuvolos@nuvolos:/files/JPE-Miller-20240145/replication-package/replication_package$ python -m uv run scons
scons: Reading SConscript files ...
scons: done reading SConscript files.
scons: Building targets ...
scons: `.` is up to date.
scons: done building targets.

(Miller) nuvolos@nuvolos:/files/JPE-Miller-20240145/replication-package/replication_package$ uv run scons -c
(Miller) nuvolos@nuvolos:/files/JPE-Miller-20240145/replication-package/replication_package$ uv run scons
scons: Reading SConscript files ...
scons: done reading SConscript files.
scons: Building targets ...
build_stata(["datastore/derived/events/regime_change.dta"], ["source/derived/01_events/03_create_regime_change.do",
"datastore/raw/vdem/orig/version10/V-Dem-CY-Full+Others-v10.dta", "datastore/raw/vdem/data/pre1900_dem.dta",
"datastore/raw/regime_change/orig/ERT.csv", "datastore/raw/regime_change/orig/vod_democratizations.csv",
"datastore/raw/regime_change/orig/DDCGdata_final.dta", "datastore/raw/crosswalks/valid_iso3_codes.dta"])
build_stata(["datastore/derived/events/coup_detat.dta"], ["source/derived/01_events/04_create_coup_detat.do",
"datastore/raw/vdem/orig/version10/V-Dem-CY-Full+Others-v10.dta"])
build_stata(["datastore/derived/events/default.dta"], ["source/derived/01_events/06_create_sovereign_default.do",
"datastore/raw/other_events/orig/reinhart_rogoff_defaults.dta", "datastore/raw/crosswalks/country_name_to_iso3.csv",
"datastore/raw/crosswalks/valid_iso3_codes.dta"])
build_stata(["datastore/derived/events/head_of_government_deaths.dta"],
["source/derived/01_events/07_create_hog_deaths.do", "datastore/raw/vdem/orig/version10/V-Dem-CY-Full+Others-v10.dta",
"datastore/raw/other_events/orig/assassinations_data.dta", "datastore/raw/other_events/orig/country_names_to_iso3.dta",
"datastore/raw/other_events/orig/HOG_death.csv", "datastore/raw/crosswalks/valid_iso3_codes.dta"])
build_stata(["datastore/derived/events/icb_crisis.dta"], ["source/derived/01_events/08_create_international_crises.do",
"datastore/raw/icb/orig/icb2v12.csv", "datastore/raw/crosswalks/icb_country_code_to_iso3.csv",
"datastore/raw/crosswalks/valid_iso3_codes.dta"])
scons: *** [datastore/derived/events/icb_crisis.dta] TargetNonexistenceError : The following target files do not exist
after build:
    datastore/derived/events/icb_crisis.dta
Traceback (most recent call last):
  File "/files/JPE-Miller-20240145/replication-package/replication_package/.venv/lib/python3.12/site-
packages/SCons/Action.py", line 1441, in execute
    result = self.execfunction(target=target, source=rsources, env=env)
              ~~~~~^~~~~~
  File "/files/JPE-Miller-20240145/replication-package/replication_package/source/lib/JMSLab/builders/build_stata.py",
line 32, in build_stata
    builder.execute_system_call()
  File "/files/JPE-Miller-20240145/replication-package/replication_package/source/lib/JMSLab/builders/build_stata.py",
line 79, in execute_system_call
    super(StataBuilder, self).execute_system_call()
  File "/files/JPE-Miller-20240145/replication-package/replication_package/source/lib/JMSLab/builders/jmslab_builder.py",
line 121, in execute_system_call
    self.check_targets()
  File "/files/JPE-Miller-20240145/replication-package/replication_package/source/lib/JMSLab/builders/jmslab_builder.py",
line 197, in check_targets
    raise TargetNonexistenceError(message)
source.lib.JMSLab._exception_classes.TargetNonexistenceError: The following target files do not exist after build:
    datastore/derived/events/icb_crisis.dta
scons: building terminated because of errors.
```

9 Findings

This section, and its subsections, should be actioned by the author.

- The author does not provide a distinct section 'Instructions to Replicators' with step by step instructions on how to reproduce the results.
- the file `source/lib/JMSLab/builders/executables.yml` had to be edited by the replicator to make sure the name of the executable corresponds to the local installation. It said StataMP when on our system it's called stata-mp. Author should either write code which fetches and uses the right name from the replicator's system, or include the change as a step in the 'Instructions to Replicators'.

- All the figures and tables were already included in the package so the code "short circuits" and builds the final pdf without recreating the tables and figures. For this reason, they must first be cleared from the package using `uv run scons -c`. Author should either clear the package, or add the clearing command as a step.
- `paper/` is stored in `source/`, not under `source/analysis/` which causes the code break documented above.

9.1 Data Preparation Code

In the data build:

- `datastore/derived/events/icb_crisis.dta` not found - refer to last error message of the Replication section for the error details.
- `source/derived/01_events/14_combine_all_events.do` failed to run. `AllData.dta` was not found. Replicator runs `!find . -name "allData.dta"` in the replication package root, which returned nothing.
- `source/derived/02_macro_political/01_create_inflation.do` failed to run. Also requires `AllData.dta`.
- `source/derived/03_assets/08_create_cashflow_growth.do` failed to run. Also requires `AllData.dta`.

9.2 Tables and Figures

In the analysis:

- All the appendix figures, except the first, are not reproduced because they are naming inconsistencies in the List of tables and programs. E.g. `04_figure_C4_cso_activity_vs_mobilizations.do` instead of `04_figure_C4_sample_period_move.do`.
- There is a produced `table_B5_adverse_probability.tex` which does not map to the appendix table B5. (screenshot below)

Table B.5: Democratizations and probability of adverse events

This table presents regressions of the form

$$\mathbb{I}_{c,t}\{\text{Event Start}\} = \alpha_c + \alpha_t + \beta_1 \mathbb{I}_{c,t}\{\text{Democratization}\} + \beta_2 \mathbb{I}_{c,t}\{\text{Autocratization}\} + \beta_3 \mathbb{I}_{c,t}\{\text{Regime Change}\} + \epsilon_{c,t}. \quad (\text{B.1})$$

Data are presented from 1900 on since the V-Dem constructed autocratization variable is only available over that sample. Standard errors are clustered by country. ***, **, * denote significance at the 1%, 5%, and 10% level, respectively.

Panel A: All countries

Dependent variable:	Adverse Event	Default	War	Financial Crisis	Recession	Market Loss
	(1)	(2)	(3)	(4)	(5)	(6)
Democratization	0.38 (0.47)	0.27 (0.29)	0.10 (0.35)	0.23 (0.30)	-0.36 (0.30)	0.09 (0.12)
Autocratization	2.09** (0.94)	1.70*** (0.59)	1.89** (0.81)	0.47 (0.47)	-0.29 (0.55)	0.38 (0.38)
Other Regime Change	3.43*** (0.88)	2.06*** (0.64)	4.01*** (0.79)	0.10 (0.42)	-0.04 (0.37)	0.15 (0.23)
Democratization obs.	371	371	371	371	371	371
R ²	0.07	0.05	0.05	0.06	0.12	0.02
Observations	18,306	18,306	18,306	18,306	18,306	18,306

Panel B: Countries with equity market during sample

Dependent variable:	Adverse Event	Default	War	Financial Crisis	Recession	Market Loss
	(1)	(2)	(3)	(4)	(5)	(6)
Democratization	0.26 (0.75)	0.08 (0.43)	0.46 (0.56)	0.12 (0.45)	-0.56 (0.53)	0.03 (0.19)
Autocratization	2.97* (1.54)	2.73*** (0.97)	1.84 (1.24)	1.21 (0.84)	-0.26 (1.06)	0.72 (0.74)
Other Regime Change	4.33*** (1.34)	2.76*** (0.99)	4.99*** (1.26)	0.34 (0.73)	0.39 (0.74)	-0.10 (0.31)
Democratization obs.	180	180	180	180	180	180
R ²	0.08	0.05	0.06	0.08	0.13	0.03
Observations	9,404	9,404	9,404	9,404	9,404	9,404

W.r.t the python scripts:

- Figure 8 cannot be produced because the corresponding `.json` file is not in the package:

```
(Miller) nuvolos@nuvolos:/files/JPE-Miller-20240145/replication-package/replication_package$ ROOT_PATH=$(pwd) python -m
source.analysis.main.figures.08_figure_8_autocratization_figure
```

Traceback (most recent call last):

```
File "<frozen runpy>", line 198, in _run_module_as_main
File "<frozen runpy>", line 88, in _run_code
File "/files/JPE-Miller-20240145/replication-
package/replication_package/source/analysis/main/figures/08_figure_8_autocratization_figure.py", line 85, in <module>
main()
File "/files/JPE-Miller-20240145/replication-
```

```

package/replication_package/source/analysis/main/figures/08_figure_8_autocratization_figure.py", line 30, in main
    m = AutocratizationModel(PARAMS_PATH)
    ~~~~~
File "/files/JPE-Miller-20240145/replication-
package/replication_package/source/derived/model/autocratization_model.py", line 21, in __init__
    with open(json_path, 'r') as f:
    ~~~~~
FileNotFoundError: [Errno 2] No such file or directory: '/files/JPE-Miller-20240145/replication-
package/replication_package/source/derived/model/baseline_params.json'
(Miller) nuvolos@nuvolos:/files/JPE-Miller-20240145/replication-package/replication_package$ find . -name
"baseline_params.json"

```

- Table 11 cannot be reproduced because it searches for `empirics_numbers.json` at the root of the package. Also `empirics_numbers.json` is not in the package.

```

(Miller) nuvolos@nuvolos:/files/JPE-Miller-20240145/replication-package/replication_package$ ROOT_PATH=$(pwd) python -m
source.analysis.main.tables.11_table_11_model_calibration
Traceback (most recent call last):
  File "<frozen runpy>", line 198, in _run_module_as_main
  File "<frozen runpy>", line 88, in _run_code
  File "/files/JPE-Miller-20240145/replication-
package/replication_package/source/analysis/main/tables/11_table_11_model_calibration.py", line 138, in <module>
    main()
  File "/files/JPE-Miller-20240145/replication-
package/replication_package/source/analysis/main/tables/11_table_11_model_calibration.py", line 44, in main
    with open(ESTIMATES_PATH, 'r') as f:
    ~~~~~
FileNotFoundError: [Errno 2] No such file or directory: '/files/JPE-Miller-20240145/replication-
package/replication_package/empirics_numbers.json'
(Miller) nuvolos@nuvolos:/files/JPE-Miller-20240145/replication-package/replication_package$ find . -name
"empirics_numbers.json"

```

- Table 12 could not be reproduced because `baseline_params.json` is not in the package.

```

(Miller) nuvolos@nuvolos:/files/JPE-Miller-20240145/replication-package/replication_package$ ROOT_PATH=$(pwd) python -m
source.analysis.main.tables.12_table_12_model_results
Traceback (most recent call last):
  File "<frozen runpy>", line 198, in _run_module_as_main
  File "<frozen runpy>", line 88, in _run_code
  File "/files/JPE-Miller-20240145/replication-
package/replication_package/source/analysis/main/tables/12_table_12_model_results.py", line 273, in <module>
    main()
  File "/files/JPE-Miller-20240145/replication-
package/replication_package/source/analysis/main/tables/12_table_12_model_results.py", line 31, in main
    m = DemocratizationModel(PARAMS_PATH)
    ~~~~~
File "/files/JPE-Miller-20240145/replication-
package/replication_package/source/derived/model/democratization_model.py", line 22, in __init__
    with open(json_path, 'r') as f:
    ~~~~~
FileNotFoundError: [Errno 2] No such file or directory: '/files/JPE-Miller-20240145/replication-
package/replication_package/source/derived/model/baseline_params.json'
(Miller) nuvolos@nuvolos:/files/JPE-Miller-20240145/replication-package/replication_package$ find . -name
"baseline_params.json"

```

9.3 In-Text Numbers

- ☒ In-text numbers not verified.
- ☐ There are no in-text numbers, or all in-text numbers stem from tables and figures.
- ☐ There are in-text numbers, but they are not identified in the code

10 Classification

- ☐ full reproduction
- ☐ full reproduction with minor issues
- ☒ partial reproduction (see above)
- ☐ not able to reproduce most or all of the results (reasons see above)

10.1 Reason for incomplete reproducibility

- ☐ None.
- ☐ **Discrepancy in output** (either figures or numbers in tables or text differ)
- ☐ **Bugs in code** that were fixable by the replicator (but should be fixed in the final deposit)
- ☐ **Code missing**, in particular if it prevented the replicator from completing the reproducibility check
 - ☐ **Data preparation code missing** should be checked if the code missing seems to be data preparation code
- ☒ **Code not functional** is more severe than a simple bug: it prevented the replicator from completing the reproducibility check
- ☒ **Software not available to replicator** may happen for a variety of reasons, but in particular (a) when the software is commercial, and the replicator does not have access to a licensed copy, or (b) the software is open-source, but a specific version required to conduct the reproducibility check is not available.
- ☐ **Insufficient time available to replicator** is applicable when (a) running the code would take weeks or more (b) running the code might take less time if sufficient compute resources were to be brought to bear, but no such resources can be accessed in a timely fashion (c) the replication package is very complex, and following all (manual and scripted) steps would take too long.
- ☒ **Data missing** is marked when data *should* be available, but was erroneously not provided, or is not accessible via the procedures described in the replication package
- ☐ **Data not available** is marked when data requires additional access steps, for instance purchase or application procedure.
- ☐ **Missing README** is marked if there is no README to guide the replicator, or the README is not in compliance with JPE requirements